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TRILOBITES

These Iguanas Got Carried Away and Ended Up 5,000 Miles From Home

Genetic evidence suggests that the reptiles somehow managed millions of years ago to make an ocean crossing from North America to Fiji.

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A Fijian crested iguana on a coconut palm on the island of Fiji in the South Pacific. Nicholas Hess

By Asher Elbein
March 17, 2025, 3:00 p.m. ET

For decades, the native iguanas of Fiji and Tonga have presented an evolutionary mystery. Every other living iguana species dwells in the Americas, from the Southwestern United States to the Caribbean and parts of South America. So how could a handful of reptilian transplants have ended up on two islands in the South Pacific, over 4,970 miles away?

“The question has definitely captured the imagination of scientists and the public alike,” said Simon G. Scarpetta, an evolutionary biologist at the University of San Francisco.

In research published Monday in the Proceedings of the National Academy of Sciences, Dr. Scarpetta and his colleagues make the case that the ancestors of Fiji's iguanas crossed on mats of floating vegetation. Such a voyage across nearly 5,000 miles of open ocean would be the longest known by a nonhuman vertebrate.

Rafting — the term scientists use for hitching a ride across oceans on uprooted trees or tangles of plants — has long been recognized as a way for small creatures on land to reach islands, said Hamish G. Spencer, an evolutionary geneticist at the University of Otago in New Zealand who was not involved in the study. Usually those are invertebrates, whose small size means they can survive a long way in an uprooted tree trunk. While examples from nonflying vertebrates are relatively rare, he added, lizards and snakes seem to be able to raft farther than mammals, perhaps because their slower metabolism allows them to fast for a long time.

Iguana species have proved adept at making shorter crossings. In 1995, Dr. Scarpetta said, scientists observed at least 15 green iguanas rafting nearly 200 miles on hurricane debris from one Caribbean island to another. And researchers have long agreed that the ancestors of the iguanas of the Galápagos Islands made the nearly 600-mile trip from South America on bobbing vegetation.

A crossing to Fiji, however, represents an almost unimaginable challenge. While some researchers suggested that the Fiji iguana's ancestors had rafted there as well, Dr. Spencer said, others pointed to the vast distances as a reason for skepticism. They countered that the iguanas were the remnant of an extinct group, one that had possibly crossed over land from the Americas to Asia or Australia, and then made the relatively easier crossing to Fiji and Tonga.

Dr. Scarpetta's team tackled the question by trying to work out when Fijian iguana species — which belong to a distinct genus, Brachylophus — split off from their closest relatives. After the team sampled the genetics from 14 living iguana species belonging to eight genuses, its analysis suggested that the Fijian species' closest living relatives were the genus Dipsosaurus, a group of desert iguanas found in the American Southwest and northwestern Mexico.

"Compared to other iguanas, both are relatively slender in body shape," Dr. Scarpetta said, "and they have some skeletal similarities as well, such as the morphology of their teeth." The team's analysis suggested that the two genuses split around 30 million and 34 million years ago.

That timing is important for a number of reasons, Dr. Scarpetta said: First, it's around the time volcanoes birthed the Fijian archipelago. Second, the cold and ice around the poles at that time would have made it impossible for any lineage of temperature-

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sensitive iguanas to make it to Asia or Australia from the Americas, and then hop to the Pacific islands. There is also no iguana fossil evidence anywhere in the Eastern Hemisphere other than Fiji or Tonga.

“North America is the most probable area of origin for iguanas in Fiji, and overwater rafting is the best supported mechanism,” Dr. Scarpetta said.

The team also argues that the ancestral desert iguanas — tolerant of heat and harsh conditions — would have been well suited for the trip. A three- to four-month crossing would have been roughly the length of Dipsosaurus’s winter hibernation, meaning the lizards could have made the voyage without starving.

And if the herbivorous reptiles rafted on a mat of vegetation, Dr. Scarpetta added, “the voyaging iguanas may even have had food on the journey.”

Dr. Spencer said, “In the past, such long-distance dispersal events appeared to be untestable stories, limited only by one’s imagination.” But while the suggestion might seem extraordinary, he said, the team makes a very convincing case. After all, the other possible origins for the Fijan iguanas would require events that might have been as unlikely or even more so, such as the extinction and utter disappearance of other iguanas along the alternative routes.

The study adds to a growing body of research, Dr. Spencer added, suggesting that “long-distance dispersal is far more important in the evolutionary history of many animal groups than had previously been appreciated.”

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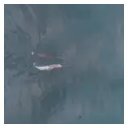
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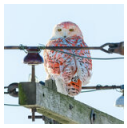
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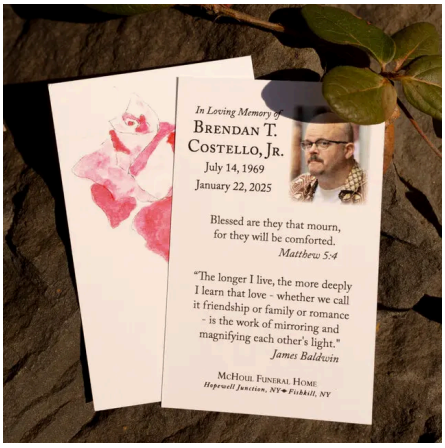
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